

- 6 Fig. 6.1 shows three equally spaced equipotential lines between two charged parallel plates. The plates are 0.12 m apart and has a potential difference of 50 V across them.

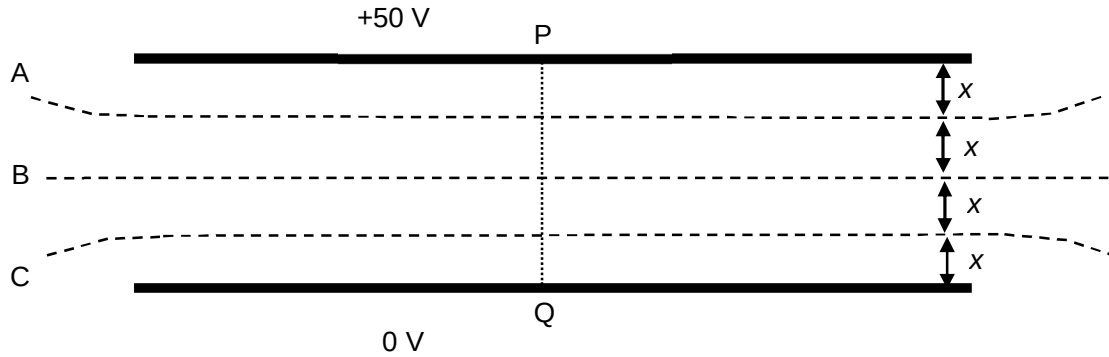


Fig. 6.1

- (a) On Fig. 6.1, draw six lines to show the electric field between the plates. [1]
- (b) On Fig. 6.2, sketch the variation of the electric potential V with x where x is the distance from P along the line PQ. Label clearly the values for the electric potential corresponding to the lines A, B and C.

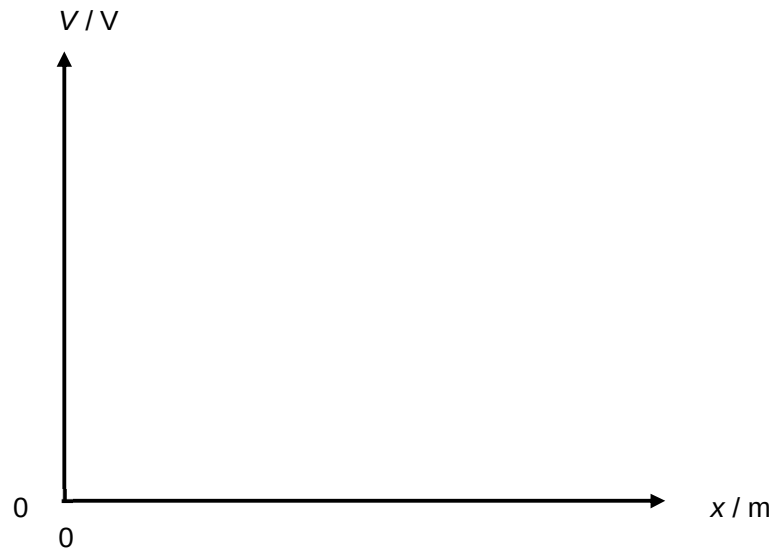


Fig. 6.2

[2]

- (c) A proton is placed at a point along the equipotential line A.

Determine the electric force acting on it.

force = N [2]

- (d) The proton is now projected vertically downwards with initial velocity of $2 \times 10^4 \text{ m s}^{-1}$ towards the lower plate.

Determine the speed at which the proton will hit the lower plate.

speed = m s^{-1} [2]

7. A piece of wire X and a negative temperature coefficient (NTC) thermistor are connected